

1. Several key differences exist between HTGR and LWR. These are highlighted in a table below

Reactor	Characteristics
HTGR	• Higher operating temperature $> 800^{\circ}\text{C}$, as high as 1000°C for VHTGR • Particle-type fuel (TRISO), two reactor types: Pebble bed and Prismatic core • Gas cooled, He primarily used as coolant, though others have been explored.
LWR	• operating temperature $\sim 300^{\circ}\text{C}$ • ceramic UO_2 fuel, Zr cladding rod-type fuel • Water-cooled reactor, with PWR and BWR styles.

2. There are 5 layers in a TRISO particle:

- a. Fuel kernel
- b. Buffer layer
- c. Inner pyrolytic carbon layer (IPyC)
- d. Silicon carbide cladding layer
- e. Outer pyrolytic carbon layer (OPyC)

The fuel kernel of course houses the fuel, typically UO_2 for the most part. Recent implementations have used UCO fuel which is 80% UO_2 and 20% UG fuel.

The buffer layer acts as a guard between the SiC/IPyC layers and fission products released from the kernel. As the kernel swells, the buffer and IPyC layers increase in stress and deteriorate due to released volatiles.

The Silicon carbide layer is the primary containment vessel/cladding, preventing release of fission products.

The IPyC/OPyC layers guard the SiC layer from volatiles and corrosion.

3. Oxide fuel kernels are easier to manufacture, as oxidation is not a concern. Carbide fuels are further more complicated to fabricate since oxidation is a concern. Oxide fuels increase the oxygen potential of the TRISO particle, leading to higher swelling and generation of CO gas. Carbide fuels typically have a lower oxygen release, leading to a significantly different swelling behavior. This is why UCO fuel has been of significant interest. The UC fuel in a UCO system first is completely oxidized before oxidation of other components of the particle, thereby restricting the generation of CO gas and fuel swelling.

4. Irradiation damage in SiC is highly impacted by fabrication method, SiC quality and the presence of impurities. Generally for SiC with minimal impurities, damage accumulation up to an irradiation temperature of $\sim 420^\circ\text{C}$ is driven by point defects. Swelling exerts pressure on the microstructure until it becomes amorphous (at the critical threshold). Point defects continue to generate, however swelling begins to decrease after this point as recombinations increase. This reduction continues until higher irradiation temperatures are reached ($> 1000^\circ\text{C}$) when void formation and void swelling begins to dominate. Saturation of void swelling is not known to occur.

5. Thermal conductivity is driven primarily via phonon transport in the material. Some degradation is observed initially until point defect saturation occurs, then the thermal conductivity remains roughly constant. However, once voids form and become dominant, the voids act as ^{primary} phonon scatterers, significantly affecting and decreasing the thermal conductivity with increasing irradiation temperature.

6. Fission product generation in the fuel is of course of primary concern. The SiC cladding is intended to prevent release, however Pd, a primary product generated in the fuel, has a corrosive effect on SiC. At higher burnups, Pd can migrate to the SiC layer and begin corroding it.

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Other fission products generated in the fuel also include Rb, I, Cs, among others. I and Cs are of primary concern and Cs ~~has~~ ^{plays} an influence on fuel chemistry at higher burnup as well.

7. Graphite is very anisotropic because the bonds between basal planes are far weaker than the bonds within the basal plane. Under a large temperature gradient this leads to cracks between "layers" of graphite. Under irradiation the bonds between the layers can actually get stronger (stronger than the van der Waals bonds (interlayer)). This drives an "increase" in G , driving a "decrease" in A . This leads to overall shrinkage of the graphite. Shrinkage will continue to happen until a threshold is reached, and then swelling in the graphite continues through end of life. This behavior is significantly influenced by creep behavior, as creep significantly influences and effectively reduces/moderates the stresses seen in the graphite due to shrinkage. Creep is enhanced under compressive forces while lower when in tension. Creep is a major benefactor to the usability of graphite in nuclear reactor systems.

8. Failure mechanisms

1. Cracking in the IPyC leading to fracture of the SiC layer.

Due to the tight bonding between the SiC and IPyC layers, as a crack forms in the IPyC, it creates a stress concentration at the SiC layer as well, leading to a radial fracture in the SiC layer.

2. IPyC / SiC layer delamination. At higher burnup, the graphite in the IPyC has a shrinking effect (before swelling), while the SiC layer is driven to thermally expand. This is typically somewhat regulated by the compressive stress from the OPyC, however high stresses still are generated at the IPyC / SiC interface.

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This leads to circumferential cracking, weakening the SiC layer, and possibly leaving it more susceptible to fission products.

3. Fuel kernel migration (Cabrera effect). Due to the stress state in the particle, potential manufacturing defects, and gas generation inside the particle, the fuel kernel position can migrate off-center closer to the cladding wall, increasing potential for corrosion and release of fission products.

4. Other failure types include cracking due to OPyC failure.

9. ZrC coating of the OPyC layer can play a significant effect on changes in the OPyC that can drive failure. ~~that~~ ZrC coated SiC also greatly reduces corrosion of the SiC from Pd. This doesn't seem to effect migration of silver though.

10. The stress state of SiC is very dynamic. Due to graphite shrinking the inside interface (at the OPyC) is in tension while the outside is in compression. This eventually will switch and invert at higher burnup after the graphite layers begin to swell instead of shrink.